

SELF-ADJOINT EXTENSION SELECTION UNDER QUADRATIC FORM ORDER IN QUANTUM MECHANICS

AMOS JAY MALEY

ABSTRACT. Self-adjoint extension is a standard mechanism by which symmetric operators in quantum mechanics are promoted to physically admissible observables. This paper proves a bounded fixed-base classification for such extension and readback procedures. The general structural result is that a carrier-preserving procedure cannot totalize a genuinely partial old operation on old data without changing operator graph, carrier, auxiliary data, language, or admissibility policy. The main application concerns closed symmetric operators. For equal nonzero deficiency indices, von Neumann theory parameterizes self-adjoint extensions by unitaries $U : \mathcal{N}_+ \rightarrow \mathcal{N}_-$. The unitary redescription group G_T acts on these parameters, and invariant parameters are exactly unitary intertwiners between the deficiency representations. If no such intertwiner exists, no invariant selection exists; if one exists, its scalar phase orbit gives distinct self-adjoint extensions. Redescription invariance alone therefore cannot select a unique nontrivial realization. The positive operator-theoretic result is an endpoint characterization in the closed quadratic-form order: for a lower semibounded symmetric operator, the Friedrichs extension is exactly the unique extension dominating every lower semibounded self-adjoint extension in that order. The Krein–von Neumann extension is the dual lower endpoint in the nonnegative case. Thus non-unique extension either has no canonical invariant selector, is determined by an explicit form-order endpoint criterion, requires boundary or variational data, or changes the operator framework. No new empirical physics is asserted.

Affiliation: Independent researcher, Pho Thong, Thailand
Corresponding author: amos@constraintsystemsinsitute.org

1. INTRODUCTION

This paper asks a bounded structural question about operator frameworks used in mathematics and mathematical physics. Once a base structure is fixed, can a genuinely internal operator create new admissible operator power on that same base? Here “internal” means carrier-preserving and determined by data fixed by the original base, rather than by witnesses, selectors, evaluators, comparison maps, completions, or ambient objects imported from outside

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that base. The question arises because many standard methods—completion, saturation, forcing, universal constructions, canonicalization, spectral or representational enrichment, and representation change—produce conclusions that are later read back to an original subject matter. The issue is not whether those methods are legitimate or fruitful. It is whether, relative to a fixed base, they amount to new same-base operator content.

The central claim is negative and deliberately narrow. On a fixed base structure equipped with an admissibility regime for partial operations, every purported carrier-preserving strengthening is either conservative on old-base-relevant content or else depends essentially on external data and is not internal to that base. In particular, a carrier-preserving extension cannot totalize an operation that was genuinely partial on old tuples. The result is therefore a boundary theorem about internality, not a global thesis that external mathematics is dispensable.

For operator theory readers, the concrete test case is the passage from a densely defined symmetric operator to a self-adjoint extension. In the non-essentially-self-adjoint case, the familiar von Neumann parameterization by a unitary map between deficiency subspaces supplies a precise load-bearing datum. The present theorem does not reprove the existence theory of self-adjoint extensions; it asks whether that datum can be selected from the original symmetric-operator base alone. The invariant-selection theorem proves that redescription-invariant candidates are exactly unitary intertwiners between the deficiency representations and that any such candidate carries an unavoidable scalar phase family of distinct extensions. Thus, in the equal nonzero deficiency case, fixed-base invariance gives no unique extension selector. The positive orthodox counterpart is stated at the level of closed quadratic forms: for a lower semibounded symmetric operator, the Friedrichs extension is characterized by a necessary and sufficient upper-endpoint condition in the form order, while the Krein–von Neumann extension is the corresponding lower endpoint in the nonnegative case. The result therefore separates failure of canonical selection from the explicit extremal form-order data that make a canonical endpoint selection possible.

The argument has three engines. First, in a partial-structure setting, if an old primitive becomes newly defined on an old tuple, then the induced old structure has changed. Such a move is not carrier-preserving in the relevant sense. Second, once the effect of a purported operator is mediated by a selector, witness, evaluator, completion boundary, comparison map, or generic object, one asks whether that datum is already fixed by the base. If it is, the operator is conservative on old-base-relevant content; if not, the operator is external. Third, the paper proves an exhaustion theorem for the possible loci of apparent internal strengthening and then applies that scheme to standard families. Thus the nontrivial content is not the isolated claim that an old partial graph cannot be changed without changing structure. The nontrivial content is the uniform reduction showing that the standard detours by which one might appear to gain new fixed-base power all reduce to

old-graph alteration, carrier change, auxiliary witness/evaluator dependence, language-only redescription, or regime change.

Fixed-Base Structural Classification of Extension Mechanisms. The general fixed-base classification used later in the paper is the following. It is a bookkeeping result for extension and readback procedures, not a substitute for the operator-theoretic endpoint theorem proved in Section 7.

Theorem 1.1 (Fixed-base classification template). *Within the fixed comparison class used here, any old-base-relevant operator-level extension or readback mechanism reduces to one of a finite set of structural loci: alteration of the old operator graph, change of carrier, introduction of evaluative or comparison data, conservative definitional expansion, or modification of admissibility policy. Consequently, no such mechanism yields new same-base operator content on the fixed base except by conservative reformulation or dependence on auxiliary data not fixed by that base.*

The remainder of the paper establishes this classification by combining rigidity of partial operations, classification of auxiliary data, and stability of the classification under composition and interaction.

Structural Completeness of Operator-Level Transformations. The classification result can be stated as a structural completeness claim: the five structural loci are not merely a convenient list of examples, but an exhaustion of the possible ways an operator-level transformation can affect a fixed base.

Theorem 1.2 (Completeness of fixed-base transformation loci). *Let $\mathcal{B}$ be a fixed base structure. Any operator-level transformation that affects values or admissible consequences on $\mathcal{B}$ must act through one of the following structural loci:*

- (i) *modification of the operator graph on existing tuples;*
- (ii) *change of carrier;*
- (iii) *introduction of evaluative or comparison structure;*
- (iv) *conservative definitional expansion;*
- (v) *modification of admissibility conditions.*

These loci are exhaustive: no operator-level transformation affecting the fixed base can arise outside this classification.

Proof sketch. An operator-level transformation can change old-base-relevant content only by changing values on old tuples, changing which tuples or objects are being compared, adding a rule for evaluating or comparing cases not determined by the base, adding only conservative notation, or changing the admissibility policy under which operations are licensed. These are exactly the five loci. Composition and iteration do not add a sixth locus, because any non-conservative effect must appear at the earliest stage where one of these changes occurs. $\square$

Non-triviality of the result. The rigidity statement alone—that old partial operations cannot be totalized without altering the underlying structure—is elementary. The non-trivial content of the present work is the demonstration that a wide range of extension and readback mechanisms, including forcing, compactness, completion, and universal constructions, are governed by the same fixed-base obstruction. The result is therefore not a restatement of rigidity, but a classification theorem for determining when such methods amount to conservative readback, auxiliary-data dependence, or changed framework relative to a fixed base.

1.1. Why the result is not merely definitional. The main theorem would be uninteresting if “internal” were simply defined to mean “conservative”. That is not the structure of the proof. Conservativity is one possible outcome of a fixed-base comparison. The paper first fixes a comparison class in which the old carrier, old primitive graphs, old admissibility profile, and old-base-relevant consequence space are the objects under comparison. It then proves two structural facts inside that class.

The first fact is rigidity: no carrier-preserving extension totalizes a genuinely partial old primitive on old tuples. This is not a relabeling of totalization as external; it is a statement about equality of induced partial graphs. The second fact is a five-locus exhaustion: any claimed old-base-relevant strengthening must alter the old graph, the carrier, the auxiliary evaluator/witness layer, the language only, or the operative admissibility policy. Only after these possibilities have been isolated do the labels “conservative” and “external” enter. The theorem is therefore a structural classification of forced cases, not a stipulative taxonomy.

A counterexample to the classification claim must accordingly do more than exhibit a useful external method. It must exhibit a carrier-preserving procedure whose old-base-relevant effect is not conservative, whose output is determined by the fixed base alone, and whose operation does not alter any of the five structural loci just listed.

1.2. Scope and roadmap. The intended contrast is between internal generation and external method. Forcing works by adjoining generic objects and passing to a richer ambient universe; metric completion passes to a completed object; saturation and compactness work via larger models or witness structures; universal constructions are posed relative to ambient comparison data [6, 7, 8, 9, 10, 11]. None of this is challenged here. The claim is only that, measured against a fixed original base, such methods are external or scope-extending unless the final readback is conservative on the old base.

Section 2 positions the paper relative to conservative extension, theory comparison, partial structures, model theory, forcing, universal constructions, and foundational methodology. Section 3 fixes the formal framework and comparison class. Section 4 proves the no-internal-totalization theorem. Section 5 proves the auxiliary-data theorem. Section 6 proves the five-locus exhaustion theorem. Section 7 develops the flagship families. Section 8 shows

that recombination does not generate a third case. Section 9 states the final theorem. Section 10 records implications for operator-theoretic frameworks in mathematical physics. Appendices record expanded family reductions, the dependency map, and submission declarations.

2. RELATION TO PRIOR WORK AND POSITIONING

This paper sits at the junction of several established literatures. Its main theorem is new in its exact form, but its proof language and case studies draw on standard work in logic and foundations. The point of this section is not to claim a single narrow ancestry. It is to make clear which literatures the manuscript enters and what it takes from each of them.

Operator theory and quantum-mechanical domains. The physics-facing application is self-adjoint extension theory for densely defined symmetric operators. Standard functional analysis treats the passage from a symmetric operator to a self-adjoint realization through closures, adjoints, deficiency subspaces, boundary conditions, and, when the deficiency indices agree, unitary maps between the deficiency spaces [15, 13]. The present paper does not alter that theory. It asks a fixed-base question about it: when the original symmetric operator and its domain are held fixed, which part of the self-adjoint realization is already determined by the old operator data, and which part is supplied by extension data or by a changed operator framework?

Conservative extension and theory comparison. One strand of the paper concerns when an apparent strengthening says nothing genuinely new about the original base. That issue belongs to the familiar study of definitions, expansions, interpretations, and conservative extensions in model theory and adjacent work on theory comparison [6, 7]. More recent discussions of definitional, categorical, and Morita equivalence sharpen the idea that two presentations may differ in signature while agreeing on what is added about their original subject matter [2]. The present manuscript does not collapse those notions into one equivalence relation. It uses them as anchors for the narrower claim that internally fixed auxiliary data yield only conservative change.

Partial structures and genuine undefinedness. The no-totalization result belongs formally to the setting of partial structures and partial algebras, where undefinedness is part of the structure rather than a temporary defect. Burmeister provides the relevant background vocabulary [4]. The present use of partiality is sharply focused: if an old primitive acquires a new value on an old tuple, then the old base itself has changed.

External methods in model theory and set theory. Several of the paper's main examples are standard external methods of logic. Model theory uses compactness, saturation, monster models, elementary extensions, and witness structures to move to richer environments [6, 7, 11]. Set theory uses forcing, generic extensions, Boolean-valued models, and absoluteness to compare what survives passage to new universes [8, 9]. The paper does not dispute those

practices. It reclassifies their fixed-base status: relative to a fixed base, they are ambient enlargements, witness imports, or conservative readbacks, not autonomous sources of new same-base operator content.

Universal constructions and ambient comparison. Category-theoretic existence by universal property is another natural neighboring literature. Universal arrows, adjunctions, limits, colimits, and free constructions are posed relative to an ambient category and comparison maps [10]. For the present paper, that ambient dependence is not an objection; it is the feature that marks the boundary between internal generation on an object alone and existence mediated by a wider framework.

Foundational methodology. Methodologically, the closest analogue is fixed-base analysis in foundations. Reverse mathematics asks which set-existence principles are needed over a chosen base theory to prove a theorem [14]. The present paper asks a different but structurally similar question: once a base structure is fixed, which operator moves are already internal to it, and which rely on additional witnesses or scope extension?

Terminological caution and novelty. The manuscript’s use of “admissibility” is local and regime-relative; it is not meant to place the paper inside admissible set theory in the sense of Barwise [3]. The novelty claim is likewise specific. The paper is not merely putting conservative extension, forcing, saturation, and universal constructions into one list. It argues that these apparently different families are governed by a common dichotomy: once the load-bearing datum is isolated, the move is conservative because that datum was already fixed by the base, or external because it was not. Coupled with graph rigidity and the five-locus exhaustion theorem, that is the unified contribution of the manuscript. The novelty lies in demonstrating that a wide range of extension and readback mechanisms—spanning model theory, forcing, completion, and universal constructions—are governed by a single structural obstruction schema, rather than requiring separate analyses.

3. FORMAL SETTING: BASE STRUCTURES AND THE FIXED COMPARISON CLASS

This section fixes the formal framework used throughout. The definitions are phrased in standard mathematical language while preserving the distinctions that drive the later proofs.

Definition 3.1 (Base structure). A *base structure* is a triple

$$\mathcal{B} = (X, \Omega, \mathcal{A}),$$

where X is a carrier, Ω is a family of partial operations

$$\omega : X^{n_\omega} \rightarrow X$$

of finite arities, and $\mathcal{A}$ is an admissibility regime specifying which primitive applications and compositions count as licensed on the carrier under discussion.

Remark 3.2. The admissibility regime may be presented by primitive domains of definition, by closure constraints on compositions, or by an explicit class of licensed constructions. The essential point is that admissibility is part of the structure under analysis, not a temporary gap to be repaired later.

Definition 3.3 (Old-base-relevant content). Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a base structure. The *old-base-relevant content* of a presentation consists of:

- (i) the carrier X under discussion;
- (ii) the graphs of the old primitive operations in Ω on tuples from X ;
- (iii) the admissibility profile induced on old primitive applications and old finite compositions;
- (iv) the consequences about the old base expressible in the old comparison language.

Old-base-relevant content therefore refers, in the minimal formal sense used here, to values of primitive operations on old tuples and consequences expressible in the original signature of the base structure. The exact old comparison language may vary by application, but it is fixed before a purported strengthening is assessed.

Definition 3.4 (Fixed-base comparison class). A purported operator, redescription, extension, or strengthening remains in the *fixed-base comparison class* of $\mathcal{B}$ when it preserves all old-base-relevant content as the object of comparison: the carrier X , the old primitive operations on old tuples, the induced old admissibility profile, and the old comparison language in which readback claims are made. A change to one of these items is an explicit scope change unless it is only a conservative definitional redescription.

Remark 3.5 (Comparison-class lock). This comparison class is not introduced to insulate the theorem from counterexamples. It records the minimum data needed for the question to be well-typed. If the carrier, the old operation graphs, the admissibility profile, or the old comparison language are changed without notice, the resulting construction may be mathematically important, but it is no longer a carrier-preserving operator on the original base in the sense under review.

Definition 3.6 (Carrier-preserving operator). Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a base structure. A purported operator F on admissible inputs from X is *carrier-preserving* if it is presented as acting on the same base carrier X rather than by openly replacing X with a quotient, completion, extension, or new ambient structure.

Definition 3.7 (Conservative action on old-base-relevant content). A purported operator or redescription on $\mathcal{B}$ acts *conservatively on old-base-relevant content* if it adds no new values for old operations on old tuples and no new consequences about the old base beyond those already fixed by the original regime and the old comparison language. In specific applications this may be sharpened to definitional extension, interdefinability, conservative extension over an old signature, or a similar standard relation.

Definition 3.8 (Admissibility-preserving redescription). An *admissibility-preserving redescription* of $\mathcal{B}$ is a change of coding, notation, presentation, or equivalent formal language that preserves the carrier under comparison, the old operation graphs, the old admissibility profile, and the old-base-relevant consequences.

Definition 3.9 (Internally fixed auxiliary datum). Let $x \in X$ be an admissible input. An auxiliary datum $a(x)$ is *internally fixed over x* if every admissibility-preserving redescription of the base that leaves the old-base-relevant content of x unchanged also leaves $a(x)$ unchanged up to normalizations already licensed by the regime.

Definition 3.10 (External parameter or witness). An auxiliary datum relevant to a purported operator on x is *external* if it is not internally fixed over x and must therefore be supplied by an extra selector, witness, evaluator, generic object, comparison map, completion rule, or ambient construction not already determined by the original base structure.

Proposition 3.11 (Internality firewall). *Throughout this paper the following principles are fixed.*

- (i) *Admissibility is structural rather than merely provisional.*
- (ii) *A carrier-preserving redescription cannot create new values for old operations on old tuples.*
- (iii) *If a purported carrier-preserving operator depends essentially on external parameters or witnesses, then it is not internal to the original base structure.*

Proof. Clause (i) is built into the role assigned to the admissibility regime. Clause (ii) is exactly the content of conservative action on old-base-relevant content in this setting. Clause (iii) follows because an operator whose effect depends on data not fixed by the base is not determined by that base alone. $\square$

Remark 3.12 (Definability is not fixed-base internality). Internality is evaluated relative to the fixed base structure. Definability in an expanded language, a forcing extension, a monster model, a completion, a quotient, or another ambient representation does not by itself constitute same-base operator content on the original base. It counts as fixed-base internal only when the resulting old-base-relevant values or consequences are determined without dependence on the added witness, evaluator, comparison map, or ambient structure.

4. RIGIDITY OF CARRIER-PRESERVING EXTENSION

We now formulate the first theorem engine. The underlying idea is simple: if an old partial operation becomes newly defined on an old tuple, then the old base structure has changed. The point of the section is to make that observation formal enough to support the later fixed-base classification.

Definition 4.1 (Weak admissible extension). Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a base structure. A *weak admissible extension* of $\mathcal{B}$ is a structure

$$\mathcal{B}^* = (X^*, \Omega^*, \mathcal{A}^*)$$

such that:

- (i) $X \subseteq X^*$;
- (ii) for each primitive $\omega \in \Omega$ there is a primitive $\omega^* \in \Omega^*$ of the same arity whose restriction to old defined tuples agrees with ω ;
- (iii) compositions of old primitives in the extension restrict to the old compositions on old admissible tuples.

This definition deliberately does *not* assume that $\text{dom}(\omega^*) \cap X^{n_\omega} = \text{dom}(\omega)$.

Definition 4.2 (Carrier-preserving extension). A weak admissible extension $\mathcal{B}^*$ of $\mathcal{B}$ is *carrier-preserving* if it is presented as remaining in the fixed-base comparison class of $\mathcal{B}$: the old carrier, old primitive structure, old admissibility profile, and old comparison language are claimed to remain the objects under discussion rather than being replaced by a quotient, completion, or new ambient domain.

Definition 4.3 (Nondegeneracy). A base structure $\mathcal{B} = (X, \Omega, \mathcal{A})$ is *non-degenerate* if at least one primitive operation in Ω has a proper, non-total domain on X .

Lemma 4.4 (Old-graph rigidity). *Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a base structure and let $\mathcal{B}^* = (X^*, \Omega^*, \mathcal{A}^*)$ be a weak admissible extension. Suppose there exist a primitive $\omega \in \Omega$ of arity n and a tuple $a \in X^n$ such that*

$$a \notin \text{dom}(\omega) \quad \text{but} \quad a \in \text{dom}(\omega^*).$$

Then the induced partial structure on the old carrier X has changed. In particular, the move is not conservative on the old base.

Proof. The graph of an old primitive operation on old tuples is part of the old structure. If a tuple that was previously outside $\text{dom}(\omega)$ becomes newly defined under ω^* , then the graph of that primitive has been enlarged on old data. That is not a mere change of notation or presentation. It is a change in the induced structure on X itself. $\square$

Lemma 4.5 (Domain fixity on the old carrier). *Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a nondegenerate base structure, and let $\mathcal{B}^*$ be a carrier-preserving extension of $\mathcal{B}$. Then for every primitive $\omega \in \Omega$ of arity n ,*

$$\text{dom}(\omega^*) \cap X^n = \text{dom}(\omega).$$

Proof. Because each old primitive is preserved by restriction, every old defined tuple remains defined, so

$$\text{dom}(\omega) \subseteq \text{dom}(\omega^*) \cap X^n.$$

Assume the inclusion is strict. Then some old tuple becomes newly defined in the extension. By Lemma 4.4, the induced partial structure on the old

carrier has changed. But a carrier-preserving extension is one that leaves the old carrier and its old induced structure in place rather than replacing it by a new regime. Strict inclusion is therefore impossible. $\square$

Theorem 4.6 (No internal totalization). *Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a nondegenerate base structure. No carrier-preserving extension of $\mathcal{B}$ totalizes a genuinely partial old operation on old tuples.*

Proof. Suppose some old primitive $\omega \in \Omega$ is partial on X but becomes total on old tuples in a carrier-preserving extension. Then $\text{dom}(\omega^*) \cap X^n = X^n$ while $\text{dom}(\omega) \subsetneq X^n$, contradicting Lemma 4.5. $\square$

Corollary 4.7 (No post hoc repair). *An old undefined primitive application cannot be made admissible later by a carrier-preserving extension of the same base structure.*

Proof. Any such repair would assign a new value to an old tuple for an old primitive operation, which is the graph change excluded by Lemma 4.5 and Theorem 4.6. $\square$

Corollary 4.8 (No deferred admissibility). *If an old tuple is outside the domain of an old operation at the stage of formation, no later carrier-preserving extension makes that old application defined on the same old data.*

Proof. Deferral changes only the timing of the same forbidden move. The old tuple would still have to become newly defined on the old carrier. $\square$

Proposition 4.9 (No definitional totalization loophole). *The no-totalization theorem is not built into the definition of extension. Weak admissible extension preserves old values where they were already defined, but it does not stipulate equality of old domains. Domain equality on old tuples is derived in Lemma 4.5 from old-graph rigidity and the fixed-base comparison claim.*

Proof. Inspect the definition of weak admissible extension: it contains no clause asserting $\text{dom}(\omega^*) \cap X^n = \text{dom}(\omega)$. That equality appears only as the conclusion of Lemma 4.5. Theorem 4.6 is then deduced from that lemma. $\square$

5. THE INTERNAL/EXTERNAL AUXILIARY-DATA THEOREM

The rigidity theorem handles direct attempts to change old operation values. Many purported internal operators do not advertise themselves as changing an old graph. Instead they introduce an auxiliary selector, witness, evaluator, completion rule, comparison map, or generic object. The question is then whether that auxiliary datum was already fixed by the original base or supplied from outside.

Definition 5.1 (Determination by fixed-base data). Let F be a purported carrier-preserving operator on admissible input $x \in X$. The old-base-relevant content of $F(x)$ is *determined by fixed-base data* if any two admissibility-preserving redescriptions that agree on the old-base-relevant content of x also agree, after licensed normalization, on the old-base-relevant content of $F(x)$.

Lemma 5.2 (Redescription test for imported data). *Suppose the old-base-relevant content of $F(x)$ varies with a datum $a(x)$. If some admissibility-preserving redescription leaves the old-base-relevant content of x fixed but changes $a(x)$ in a way that changes the old-base-relevant content of $F(x)$, then $F(x)$ is not determined by fixed-base data.*

Proof. This is the contrapositive of fixed-base determination. Agreement on the old-base-relevant input content must force agreement on the old-base-relevant output content. If the output changes only because a datum not fixed by the base changes, then the base alone did not determine the output. $\square$

Theorem 5.3 (Auxiliary-data dichotomy). *Let F be a carrier-preserving operator on admissible input $x \in X$ for a base structure $\mathcal{B} = (X, \Omega, \mathcal{A})$. Suppose that the old-base-relevant content of $F(x)$ is mediated, up to conservative redescription, by x together with an auxiliary datum $a(x)$. Then:*

- (a) *if $a(x)$ is internally fixed over x , then F acts conservatively on the old-base-relevant content of x ;*
- (b) *if $a(x)$ is external, then F is not internal to the base structure.*

Proof. Assume first that $a(x)$ is internally fixed over x . Every admissibility-preserving redescription of the base that leaves the relevant content of x unchanged also leaves $a(x)$ unchanged up to licensed normalization. Thus $a(x)$ contributes no old-base-relevant distinction beyond what was already fixed by the base together with x . The effect of F is only to reorganize or read off structure that was already there, so F is conservative on old-base-relevant content.

Assume instead that $a(x)$ is external. Then the effect of F depends essentially on a datum not determined by the old base together with x . By Proposition 3.11, such dependence is incompatible with internality relative to the original base structure. $\square$

Corollary 5.4 (No internalization by relabeling). *If an external datum is redescribed as latent, semantic, structural, canonical, or definable in an enlarged setting without becoming internally fixed over the original input, its status does not change. Any operator using it remains external to the base.*

Proof. A change of description does not make an unfixed datum fixed. If the relevant effect still depends on data not determined by the original base, then the operator still depends on an external parameter or witness. $\square$

Remark 5.5. The dichotomy is not a psychological or philosophical distinction about how a mathematician found the construction. It is an invariance test. A datum is internal only when it is fixed under the admissibility-preserving redescrptions that leave the old-base-relevant input content unchanged.

6. EXHAUSTION OF APPARENT INTERNAL EXTENSION MECHANISMS

Theorem 4.6 blocks direct totalization of old partial operations, and Theorem 5.3 classifies operators by the status of the data they require. The

next task is to show that the main kinds of apparent internal strengthening are exhausted by a structural list of loci rather than by an informal list of examples.

Definition 6.1 (Old-base-relevant strengthening). A purported procedure τ is an *old-base-relevant strengthening* of $\mathcal{B}$ if it is presented as carrier-preserving and as adding some value, operator effect, or consequence concerning the old base that was not already fixed by the original old-base-relevant content.

Definition 6.2 (Five structural loci). A purported old-base-relevant strengthening can affect the fixed comparison only through the following loci:

- (I) the old operation graph on old tuples;
- (II) the carrier or identity of the object under discussion;
- (III) an auxiliary evaluator, witness, selector, completion boundary, comparison map, generic object, or semantic device;
- (IV) the descriptive language alone;
- (V) the operative admissibility policy or operator family.

Theorem 6.3 (Five-locus exhaustion). *Every apparent internal strengthening of a fixed base structure falls under at least one of the five loci in Definition 6.2.*

Proof. Let τ be a purported strengthening. If it makes an old primitive newly defined on an old tuple, changes an old value, or alters an old primitive graph, it falls under locus (I). If the appearance of strength comes because the object under discussion has become a quotient, completion, extension, replacement, or otherwise different carrier, it falls under locus (II). If previously unresolved or non-determined cases are decided by an added evaluator, witness, selector, comparison map, semantic oracle, completion boundary, generic object, or analogous device, it falls under locus (III). If none of these changes occurs and only the descriptive apparatus changes, it falls under locus (IV). Any remaining case must change which constructions count as licensed, which operators are available, or which admissibility policy governs the base; that is locus (V). The coverage is structural rather than taxonomic. An old-base-relevant operator effect is determined by its value assignment on old tuples, the domain on which those assignments are licensed, the carrier to which those assignments belong, the comparison or evaluation rule used to decide previously non-determined cases, the language in which the effect is stated, and the regime constraints determining which moves are licensed. These are exactly the five loci above: old graph, carrier, auxiliary evaluator or witness layer, language, and operative admissibility policy. These five loci exhaust all ways an operator can affect old-base-relevant content, since such content is determined by value assignment, domain of definition, comparison rules, and admissibility constraints. Encoding tricks, hybrid constructions, and diagonal packages can mask which locus is being used, but they cannot create an additional locus of fixed-base effect.

These alternatives are not asserted to be disjoint. They are asserted to cover all ways in which a purported strengthening can enter the fixed-base comparison. $\square$

Proposition 6.4 (Disposition of the five loci). *Relative to the framework of Section 3, each locus in Definition 6.2 is blocked or reveals that no genuinely internal strengthening has occurred.*

- (I) *Old-graph enlargement is impossible by Theorem 4.6.*
- (II) *Carrier change or quotient passage is not carrier-preserving.*
- (III) *An auxiliary evaluator or witness is conservative when internally fixed and external otherwise, by Theorem 5.3.*
- (IV) *Pure language change is conservative definitional redescription.*
- (V) *A new operator family or admissibility policy is an open change of regime.*

Proof. Clause (I) is exactly the rigidity theorem. Clause (II) restates the meaning of carrier preservation. Clause (III) is the auxiliary-data dichotomy. Clause (IV) follows from the definition of conservative action on old-base-relevant content. Clause (V) changes the very regime relative to which internality was being measured. $\square$

Corollary 6.5 (Closure of internal totalization). *No apparent internal strengthening survives as a genuine carrier-preserving totalization of old partial operations on old tuples.*

Proof. Any such strengthening falls under one of the five loci by Theorem 6.3. Proposition 6.4 then blocks it or classifies it as conservative or scope-changing. $\square$

Remark 6.6 (Burden for counterexamples). A counterexample to the exhaustion theorem must specify a carrier-preserving old-base-relevant strengthening and identify where its effect enters the comparison without altering the old graph, changing the carrier, importing an auxiliary datum, merely changing language, or changing the operative regime.

7. FLAGSHIP CASE STUDIES

Table 1 records the fixed-base classification of the principal families. The subsections justify the final column.

TABLE 1. Fixed-base classification of the flagship families

Family	Load-bearing datum or move	Fixed-base classification
Selection, canonical representatives, quotienting	Representative rule, normalization criterion, or quotient passage	Conservative only if the rule is internally fixed; otherwise external. Quotient passage changes carrier.
Completion and evaluative closure	Completion object, embedding, boundary, or evaluator	External by carrier change unless only an internally fixed readback is used; otherwise external.
Compactness, saturation, model-theoretic existence	Larger model, realized type, witness structure, or existence theorem	Conservative only in the definable/readback case; otherwise external ambient enlargement or witness import.
Forcing, genericity, absoluteness	Generic object, forcing extension, Boolean-valued surrogate, or transfer criterion	Generic output is external; absoluteness/readback is conservative on old-base-relevant content when it no longer depends on the generic witness.
Universal constructions	Ambient category, comparison maps, universal witness object	Conservative only if the relevant witness/readback is already fixed; otherwise ambiently external and often carrier-changing.
Self-adjoint extensions	Boundary conditions, deficiency-space data, extension parameter, or new closed operator domain	Conservative only on old-domain readback; otherwise external auxiliary data or explicit change of operator framework.

7.1. Selection, canonical representatives, and quotienting. Selection and canonicalization are clean test cases for the auxiliary-data theorem. A base structure may determine an equivalence relation, a normalization target, or a family of interchangeable witnesses. The fixed-base question is whether it also determines a distinguished representative, or whether that representative is chosen by an additional ranking, coding, ordering, or normalization rule.

Proposition 7.1 (Canonicalization, selection, and quotienting). *Let $D \subseteq X$ be an admissible domain for $\mathcal{B} = (X, \Omega, \mathcal{A})$, and let $\sim$ be an equivalence relation on D invariant under admissibility-preserving redescription. Suppose a purported carrier-preserving operator $N : D \rightarrow D$ satisfies*

$$N(x) \sim x \quad \text{and} \quad x \sim y \Rightarrow N(x) = N(y).$$

Then the following trichotomy governs the case.

- (a) *If the representative rule determining $N(x)$ is internally fixed over x , then N is conservative on the old-base-relevant content of x .*

- (b) *If that rule is not internally fixed, then N depends on an external selector, ranking, code, or normalization datum and is not internal to $\mathcal{B}$.*
- (c) *If the maneuver is really passage from D to the quotient $D/\sim$, then the carrier has changed and the move is not carrier-preserving in the first place.*

Proof. If N is a map from D to D , its effect is mediated by the rule choosing which representative counts as canonical in each equivalence class. That rule is the relevant auxiliary datum. If it is internally fixed, Theorem 5.3 yields conservativity. If it is not internally fixed, the construction depends on external choice data. If the construction instead replaces each element by its equivalence class, then the object under discussion is $D/\sim$ rather than D , so the move falls under locus (II) of Definition 6.2. $\square$

Example 7.2. Choosing the least code, the lexicographically first form, or a minimal-complexity representative is internal only relative to a base that already fixes the relevant order, coding, or complexity measure. Otherwise the apparent canonicity is carried by imported structure.

7.2. Completion and evaluative closure. Completion constructions often feel canonical while still operating by passage to a richer ambient object. In metric, order-theoretic, algebraic, and semantic settings, completion typically works either by adjoining new points, classes, or ideal objects, or by adding a rule that evaluates old data against a richer completion boundary.

Proposition 7.3 (Completion and evaluative closure). *Let C be a purported completion operator applied to an admissible object A over a fixed base structure. Then exactly one of the following happens.*

- (a) *$C(A)$ is genuinely a richer object—for example, a completion, closure object, or ideal enlargement with new points, classes, or totalizations—in which case the move is not carrier-preserving.*
- (b) *The only old-base-relevant effect of the completion is mediated by an internally fixed comparison, embedding, or evaluative rule, in which case the move is conservative on the old base.*
- (c) *The readback from the completion depends on a comparison, embedding, boundary, or evaluator not internally fixed by the original base, in which case the move is external.*

Proof. If the completion produces a genuinely new object, then the carrier under discussion has changed. Suppose instead that completion is used only to obtain old-base-relevant consequences. The old-base effect is then governed by the embedding, evaluator, or completion boundary used for readback. If that datum is internally fixed, Theorem 5.3 gives conservativity. If not, the completion works by importing the additional evaluative structure on which the result depends. $\square$

Example 7.4. The passage from $\mathbb{Q}$ to $\mathbb{R}$ is mathematically canonical, but it is still passage to a richer object rather than an internal operator on the

original carrier $\mathbb{Q}$. Statements about rationals justified by working in $\mathbb{R}$ are, from the fixed-base standpoint, either conservative readbacks or consequences that still depend on an external ambient enlargement.

Remark 7.5. Uniqueness up to isomorphism or isometry controls ambiguity in the richer ambient object; it does not turn that richer object into a same-carrier operator on the original base.

7.3. Compactness, saturation, and model-theoretic existence. Model theory supplies mathematically fruitful external methods: compactness arguments, saturated and monster models, elementary extensions, and existence proofs by witness structures [6, 7, 11]. The present claim is not that those methods are dispensable. It is that, from the standpoint of a fixed base structure, they do not generate new same-base operator content unless the relevant witness was already fixed. The classification does not deny that compactness, completeness, saturation, or related principles are internal to a given logical system; it asserts only that, relative to a fixed base structure, their operational effect either depends on external witnesses or yields conservative consequences.

Proposition 7.6 (Compactness, saturation, and model-theoretic existence). *Let M be a structure, and let F be a purported operator whose effect is justified by compactness, saturation, a monster-model argument, or a model-existence theorem. Then one of the following holds.*

- (a) *The relevant witness, realization, or map was already definably fixed on the old base, in which case F is conservative on the old-base-relevant content of M .*
- (b) *The construction proceeds by passing to a larger model, adjoining new realizing elements, invoking a witness structure, or using an existence theorem whose witness is not part of M , in which case F is external to the original base.*

In particular, none of these model-theoretic methods yields new same-base operator content on a fixed old carrier except in the conservative readback case.

Proof. For type realization and saturation, if the purported operator produces a realization already available in M , then nothing new has been generated. If the realization is not already present, then some new witness element or larger ambient model is required. The same analysis applies to saturated envelopes, monster models, and compactness arguments. A compactness argument that establishes the existence of a model or element does so by producing or appealing to a witness structure not identical with the original base. If the only conclusion read back to M is an old-language consequence independent of that witness as content, then the use of the external witness is conservative on old-base-relevant content. Otherwise the load-bearing datum is external and Theorem 5.3 applies. $\square$

7.4. Forcing, genericity, and absoluteness. Forcing is a case where scope discipline is essential. The method is one of the great successes of modern logic, and much of its power comes precisely from controlled passage to richer ambient universes [8, 9]. The fixed-base diagnosis is correspondingly sharp: a forcing extension is an external move, while readback by absoluteness or invariance is conservative on the old base rather than internally generative.

Proposition 7.7 (Forcing, genericity, and absoluteness). *Let M be a ground model or other fixed set-theoretic base, and let F be a purported forcing-based operator.*

- (a) *If F works by adjoining a generic filter, generic real, or forcing extension $M[G]$, then it is external to M by imported witness or ambient enlargement.*
- (b) *If F uses forcing only to transfer a statement about old parameters by absoluteness, invariance, or ground-model forcing analysis, then its effect on old-base-relevant content is conservative rather than a new internal operator on M .*

Hence forcing and genericity do not provide new same-base operator content on the original base model.

Proof. In the first case, the generic object G is not part of the original base and the passage from M to $M[G]$ is a richer ambient universe. In the second case, the conclusion concerns only old parameters or old-base-relevant statements. Forcing then functions as a proof method or transfer method. Boolean-valued and algebraic presentations do not change the classification: they either redescribe already fixed information conservatively or shift the work onto additional evaluative or completion structure, already covered by Proposition 7.3 and Theorem 5.3. $\square$

Example 7.8. A Cohen real added over a ground model is not an internal object of that ground model. By contrast, an absoluteness argument that transfers a statement about old natural numbers back to the ground model does not thereby create a new internal operator on the ground model; it certifies old-base-relevant content by an external route.

7.5. Universal constructions and existence by universal property. Universal constructions are often canonical in a categorical sense. Their canonicity is mediated by an ambient category, a class of comparison maps, and a universal condition imposed across that ambient framework [10]. The fixed-base question is whether the relevant universal witness is already fixed by the old base or supplied through ambient categorical data.

Proposition 7.9 (Universal constructions and existence by universal property). *Let A be an object presented by a fixed base structure, and suppose a purported operator assigns to A an object $U(A)$ by a universal property in an ambient category $\mathcal{C}$. Then the following distinction is exhaustive.*

- (a) *If the universal witness and its comparison maps are already fixed, relative to the chosen ambient framework, strongly enough that the only effect on old-base-relevant content is to redescribe what was already determined, then the move is conservative on the old base.*
- (b) *Otherwise the construction depends essentially on ambient categorical data or on a genuinely new witness object $U(A)$, and it is external to the original base.*

Existence by universal property does not by itself furnish new same-base operator content on A alone.

Proof. A universal property is not posed with respect to A alone. It quantifies over a wider ambient category, comparison morphisms, and a condition singling out a witness object. If all of that structure is already fixed tightly enough that the only resulting statement about the original base is a reformulation of information determined there, the move is conservative. In the substantive case, the universal witness or comparison data do real work and are external to the base alone. $\square$

7.6. Self-adjoint extensions of symmetric operators. Self-adjoint extension theory provides a concrete operator-theoretic instance of the fixed-base distinction. Let $\mathcal{H}$ be a Hilbert space and let

$$T : \mathcal{D}(T) \subseteq \mathcal{H} \longrightarrow \mathcal{H}$$

be a densely defined symmetric operator. In quantum mechanics such an operator may represent a formally symmetric observable whose physical realization requires a self-adjoint operator. A self-adjoint extension $\tilde{T}$ of T satisfies

$$T \subseteq \tilde{T}, \quad \mathcal{D}(T) \subseteq \mathcal{D}(\tilde{T}), \quad \tilde{T}\psi = T\psi \quad (\psi \in \mathcal{D}(T)).$$

The fixed-base question is not whether self-adjoint extensions are legitimate. They are indispensable. The question is whether passage from T to $\tilde{T}$ creates new operator power internal to the original operator base [15, 13].

Definition 7.10 (Operator base associated with a symmetric operator). Let $T : \mathcal{D}(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a densely defined symmetric operator. The fixed operator base associated with T is

$$\mathcal{B}_T = (\mathcal{H}, \{T\}, \mathcal{A}_T),$$

where T is treated as a partial operator on the carrier $\mathcal{H}$, and $\mathcal{A}_T$ licenses precisely the original domain $\mathcal{D}(T)$ together with the operator applications and finite compositions already determined by the original operator framework.

Theorem 7.11 (Self-adjoint extension boundary theorem). *Let $T : \mathcal{D}(T) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ be a densely defined symmetric operator and let $\tilde{T}$ be a self-adjoint extension of T . Relative to the fixed operator base $\mathcal{B}_T$, exactly one of the following holds.*

- (i) *The extension is used only to recover statements about the original domain and original operator graph already determined by $\mathcal{B}_T$. In this case the old-base readback is conservative.*
- (ii) *The extension defines $\tilde{T}\psi$ for some $\psi \in \mathcal{D}(\tilde{T}) \setminus \mathcal{D}(T)$ while the same carrier and old operator are still being treated as the fixed base. In this case the old partial operator has been enlarged on old carrier elements, so the move is not an internal carrier-preserving strengthening of $\mathcal{B}_T$.*
- (iii) *The extension is selected by boundary conditions, deficiency-space data, a unitary map between deficiency subspaces, a boundary triple, or another extension parameter not fixed by $\mathcal{B}_T$. In this case the construction depends on auxiliary data external to the original operator base.*
- (iv) *The passage from T to $\tilde{T}$ is treated as replacement of the original operator framework by a new self-adjoint operator framework. In this case the move is an explicit change of operator base.*

Consequently, self-adjoint extension theory does not provide a counterexample to the no-internal-totalization theorem. It exemplifies the theorem: extension is either conservative on the original operator content, externally parameterized, or an explicit change of operator framework.

Proof. If $\tilde{T}$ is used only to prove or reorganize statements whose old-domain content is already fixed by T on $\mathcal{D}(T)$, then no new old-base-relevant operator value has been added, so the readback is conservative.

Suppose instead that the same carrier $\mathcal{H}$ and the old operator T remain the objects under comparison, while $\tilde{T}$ assigns a value to some vector

$$\psi \in \mathcal{D}(\tilde{T}) \setminus \mathcal{D}(T).$$

Then an originally undefined application of the old partial operator has become defined on an old carrier element. This is exactly the old-graph enlargement excluded by Theorem 4.6. Thus the move cannot be an internal carrier-preserving strengthening of the fixed base.

In the usual non-essentially-self-adjoint case, self-adjoint extensions are distinguished by additional extension data, such as boundary conditions, deficiency-space choices, or equivalent parameterizations. If those data are not already fixed by the original operator base, the resulting operator depends on auxiliary information external to that base, and Theorem 5.3 applies.

Finally, if $\tilde{T}$ is taken as the operative physical observable in place of T , then the admissible domain and operator framework have changed. That is a legitimate operator-theoretic move, but it is a change of base rather than a generation of new same-base operator content on $\mathcal{B}_T$. $\square$

Theorem 7.12 (Invariant deficiency-parameter selection theorem). *Let T be a densely defined closed symmetric operator on $\mathcal{H}$, and set*

$$\mathcal{N}_+ = \ker(T^* - iI), \quad \mathcal{N}_- = \ker(T^* + iI).$$

Assume the deficiency indices are equal and nonzero. Let

$$\mathfrak{U}(T) = \{U : \mathcal{N}_+ \rightarrow \mathcal{N}_- : U \text{ unitary}\}$$

be the von Neumann parameter space of self-adjoint extensions of T . Let G_T be the unitary redescription group of the fixed symmetric-operator base:

$$G_T = \{W \in \mathcal{U}(\mathcal{H}) : W\mathcal{D}(T) = \mathcal{D}(T), WT\psi = TW\psi \text{ for all } \psi \in \mathcal{D}(T)\}.$$

Then every $W \in G_T$ restricts to unitaries

$$W_+ : \mathcal{N}_+ \rightarrow \mathcal{N}_+, \quad W_- : \mathcal{N}_- \rightarrow \mathcal{N}_-,$$

and acts on extension parameters by

$$U \mapsto W_- U W_+^{-1}.$$

Consequently, any self-adjoint extension selection rule that is internal to $\mathcal{B}_T$ and invariant under admissibility-preserving redescriptions of the fixed operator base must select a parameter $U \in \mathfrak{U}(T)$ satisfying

$$W_- U W_+^{-1} = U \quad \text{for every } W \in G_T.$$

Equivalently, internal extension selection is possible only through a G_T -fixed unitary in the deficiency-parameter space. If the action of G_T on $\mathfrak{U}(T)$ has no fixed point, no admissibility-invariant internal selection of a self-adjoint extension exists. If a selected parameter is not such a fixed point, the selection necessarily depends on auxiliary extension data external to the original symmetric-operator base.

Proof. Since $W \in G_T$ unitarily preserves $\mathcal{D}(T)$ and commutes with T on that domain, it implements the same symmetric operator in a unitarily redescription presentation of the fixed base. Hence

$$W T W^{-1} = T$$

on $\mathcal{D}(T)$, and therefore $W T^* W^{-1} = T^*$. It follows that

$$\begin{aligned} \eta_+ \in \ker(T^* - iI) &\Rightarrow W\eta_+ \in \ker(T^* - iI), \\ \eta_- \in \ker(T^* + iI) &\Rightarrow W\eta_- \in \ker(T^* + iI), \end{aligned}$$

so W restricts to unitaries W_+ and W_- on the two deficiency subspaces.

By the von Neumann extension theorem, self-adjoint extensions of T are parameterized by unitaries $U : \mathcal{N}_+ \rightarrow \mathcal{N}_-$, with domains

$$\mathcal{D}(T_U) = \mathcal{D}(T) \dot{+} \{\eta_+ + U\eta_+ : \eta_+ \in \mathcal{N}_+\}.$$

Conjugating this domain by a base redescription $W \in G_T$ gives

$$W\mathcal{D}(T_U) = \mathcal{D}(T) \dot{+} \{W_+\eta_+ + W_-U\eta_+ : \eta_+ \in \mathcal{N}_+\}.$$

Writing $\zeta_+ = W_+\eta_+$, this becomes

$$\mathcal{D}(T) \dot{+} \{\zeta_+ + W_-UW_+^{-1}\zeta_+ : \zeta_+ \in \mathcal{N}_+\},$$

which is precisely the domain of $T_{W_-UW_+^{-1}}$. Thus the induced action on extension parameters is $U \mapsto W_-UW_+^{-1}$.

An internal selection rule relative to $\mathcal{B}_T$ cannot distinguish between admissibility-preserving redescrptions that leave the old symmetric operator base unchanged. Therefore its selected extension parameter must be invariant under the above action of every $W \in G_T$. This gives the fixed-point condition. If no such fixed point exists, an invariant internal selection rule cannot exist. If a selected parameter fails the fixed-point condition, the selection has used information not fixed by the original operator base, so it falls under the auxiliary-data alternative of Theorem 5.3. $\square$

Theorem 7.13 (Invariant selection constraint for self-adjoint extensions). *Under the hypotheses and notation of Theorem 7.12, let*

$$\rho_+ : G_T \rightarrow \mathcal{U}(\mathcal{N}_+), \quad \rho_- : G_T \rightarrow \mathcal{U}(\mathcal{N}_-)$$

be the deficiency representations obtained by restriction, and let

$$\mathfrak{U}(T)^{G_T} = \{U \in \mathfrak{U}(T) : \rho_-(W)U = U\rho_+(W) \text{ for every } W \in G_T\}.$$

Then the following alternatives are exhaustive.

- (i) *The admissibility-invariant candidate parameters are precisely the unitary intertwiners from ρ_+ to ρ_- :*

$$\mathfrak{U}(T)^{G_T} = \{U \in \mathfrak{U}(T) : U\rho_+(W) = \rho_-(W)U \text{ for every } W \in G_T\}.$$

Hence, if the deficiency representations admit no unitary intertwiner, no admissibility-invariant internal selection of a self-adjoint extension exists.

- (ii) *If $U \in \mathfrak{U}(T)^{G_T}$, then $e^{i\theta}U \in \mathfrak{U}(T)^{G_T}$ for every $\theta \in \mathbb{R}$. Since $\mathcal{N}_+ \neq \{0\}$, the extensions $T_{e^{i\theta}U}$ are pairwise distinct for distinct phases modulo 2π .*

Consequently, in the nonzero-deficiency case, redescription invariance of the fixed symmetric-operator base never by itself singles out a unique self-adjoint extension. It yields either no invariant extension parameter or a nontrivial phase family of invariant parameters. A unique selected self-adjoint realization therefore requires additional boundary, phase, normalization, comparison, variational, or physical/evaluative data already included in the operator base; if such data are not included, the selection is external auxiliary data, and if the selected realization is adopted as the operative observable, the operator framework has changed.

Proof. The fixed-point condition of Theorem 7.12 is

$$W_- U W_+^{-1} = U \quad (W \in G_T),$$

which is equivalent to

$$\rho_-(W)U = U\rho_+(W) \quad (W \in G_T).$$

Thus fixed parameters are exactly unitary intertwiners between the two restricted deficiency representations. If no such unitary intertwiner exists, then the fixed-point set is empty and no invariant internal extension parameter can be selected.

If U is such an intertwiner and $|\lambda| = 1$, then λU is also a unitary intertwiner. Write $\lambda = e^{i\theta}$. It remains to note that distinct phases give distinct self-adjoint extensions. By von Neumann's domain formula,

$$\mathcal{D}(T_U) = \mathcal{D}(T) \dot{+} \{\eta + U\eta : \eta \in \mathcal{N}_+\}.$$

If $\mathcal{D}(T_U) = \mathcal{D}(T_{\lambda U})$, then for every $\eta \in \mathcal{N}_+$ there is $\zeta \in \mathcal{N}_+$ such that

$$\eta + U\eta - (\zeta + \lambda U\zeta) \in \mathcal{D}(T).$$

The von Neumann decomposition of $\mathcal{D}(T^*)$ is direct, so the $\mathcal{N}_+$ - and $\mathcal{N}_-$ -components must agree. Hence $\eta = \zeta$ and $(1 - \lambda)U\eta = 0$. Since U is injective and $\mathcal{N}_+ \neq \{0\}$, this forces $\lambda = 1$. Therefore $T_{\lambda U} \neq T_U$ whenever $\lambda \neq 1$.

The final statement follows because a deterministic unique selection cannot be obtained from redescription invariance alone: the invariant candidate set is empty or contains the entire phase orbit just described. Choosing one element of a nontrivial phase orbit requires a further selector, boundary condition, phase convention, normalization rule, comparison datum, variational principle, or physical/evaluative criterion. By Theorem 5.3, such a datum is internal only if it was already fixed by the original operator base; otherwise it is external auxiliary data. $\square$

Definition 7.14 (Semibounded extension class and closed-form order). Let T be a densely defined closed symmetric operator on a Hilbert space $\mathcal{H}$ with $T \geq mI$. Let

$$\mathcal{E}_m(T) = \{A = A^* : T \subseteq A, A \geq mI\}$$

be the set of lower semibounded self-adjoint extensions with the same lower bound. For $A \in \mathcal{E}_m(T)$, let q_A denote the closed nonnegative quadratic form associated with $A - mI$. For $A, B \in \mathcal{E}_m(T)$, write

$$A \preceq_q B$$

if

$$\mathcal{D}(q_B) \subseteq \mathcal{D}(q_A), \quad q_A[\psi] \leq q_B[\psi] \quad (\psi \in \mathcal{D}(q_B) = \mathcal{D}(q_A) \cap \mathcal{D}(q_B)).$$

This is the standard closed-form order: the larger element has the smaller form domain and the larger form value on that domain. The order is anti-symmetric by the first representation theorem for closed semibounded forms. Throughout this subsection, every order comparison is a comparison of the closed semibounded quadratic forms associated with semibounded self-adjoint extensions.

A self-adjoint extension selector on the unitary orbit of T is a rule S assigning to every unitary copy $T' = WTW^{-1}$ an element $S(T') \in \mathcal{E}_m(T')$. The selector is *unitarily covariant* if

$$S(WTW^{-1}) = WS(T)W^{-1}$$

for every unitary W carrying T to WTW^{-1} on $W\mathcal{D}(T)$. It is *form-dominant* if, for every unitary copy T' and every $A \in \mathcal{E}_m(T')$,

$$A \preceq_q S(T').$$

Thus form dominance is the endpoint condition in the closed quadratic-form order; it is not an appeal to coordinates, boundary charts, or presentation data.

Theorem 7.15 (Friedrichs endpoint characterization theorem). *Let T be a densely defined closed symmetric operator on a Hilbert space $\mathcal{H}$, and suppose that $T \geq mI$. Let T_F be the Friedrichs extension of T . For $B \in \mathcal{E}_m(T)$, the following are equivalent:*

- (i) $B = T_F$;
- (ii) $A \preceq_q B$ for every $A \in \mathcal{E}_m(T)$.

Equivalently, for any self-adjoint extension selector S on the unitary-equivalence class of T ,

$$S(T) = T_F \quad \text{if and only if} \quad S \text{ is form-dominant at } T.$$

Moreover the Friedrichs assignment is unitarily covariant:

$$(WTW^{-1})_F = WT_FW^{-1}$$

for every unitary equivalence W . Thus the upper endpoint in the closed quadratic-form order is a necessary and sufficient operator-theoretic criterion for Friedrichs selection; no boundary triple, deficiency-space unitary, phase convention, coordinate choice, or physical/evaluative rule is used once that endpoint condition is imposed.

Proof. The shifted form

$$q_T[\psi, \varphi] = \langle (T - mI)\psi, \varphi \rangle, \quad \psi, \varphi \in \mathcal{D}(T),$$

is nonnegative and closable. Its closure $\bar{q}_T$ is a closed nonnegative form. By the first representation theorem there is a unique nonnegative self-adjoint operator A_F whose closed form is $\bar{q}_T$, and $T_F = A_F + mI$ is the Friedrichs extension of T .

The classical extremal characterization of the Friedrichs extension says that T_F is the greatest element of $\mathcal{E}_m(T)$ in the closed-form order: for every $A \in \mathcal{E}_m(T)$,

$$A \preceq_q T_F.$$

Equivalently, among closed semibounded extension forms of $T - mI$, the Friedrichs form is the maximal closed form extending the original form [12, 13, 1]. This proves that $B = T_F$ implies $A \preceq_q B$ for every $A \in \mathcal{E}_m(T)$.

Conversely, suppose $B \in \mathcal{E}_m(T)$ satisfies $A \preceq_q B$ for every $A \in \mathcal{E}_m(T)$. Applying this to $A = T_F$ gives $T_F \preceq_q B$. The maximality of T_F , applied to the extension B , gives $B \preceq_q T_F$. The closed-form order is antisymmetric on semibounded self-adjoint operators: mutual domination gives equality of form domains and equality of closed forms, and the first representation theorem then gives equality of the associated operators. Hence $B = T_F$. This proves the stated equivalence.

The selector equivalence follows by applying the previous equivalence to $B = S(T)$. Finally, if W implements a unitary equivalence from T to

$T' = WTW^{-1}$, then unitary conjugacy carries the form of $T - mI$ to the corresponding form for $T' - mI$ and carries closed-form order to closed-form order. Thus WT_FW^{-1} is the upper endpoint of $\mathcal{E}_m(T')$. By the just-proved equivalence, it is $(T')_F$. $\square$

When $T \geq 0$, the Krein–von Neumann extension occupies the opposite lower endpoint of the same closed-form order. The Krein–von Neumann extension is the minimal element under this order and therefore does not satisfy the form-dominance condition unless the Friedrichs and Krein endpoints coincide. It is therefore excluded by the upper-endpoint/form-dominance condition in Theorem 7.15 in the nontrivial case.

Lemma 7.16 (Sharpness of the endpoint characterization hypotheses). *The hypotheses in Theorem 7.15 have the following precise roles.*

- (i) *Form dominance cannot be omitted if Friedrichs selection is to be characterized by the upper endpoint. If $T \geq 0$ has distinct Friedrichs and Krein–von Neumann extensions, then*

$$S_K(WTW^{-1}) = WT_KW^{-1}$$

defines a unitarily covariant selector on the unitary orbit of T , but $S_K(T) = T_K \neq T_F$.

- (ii) *Unitary covariance is not needed for the pointwise equality once form dominance is imposed, because the greatest element of $\mathcal{E}_m(T)$ is unique. It is needed for the rule to be a representation-independent operator-theoretic selection principle: without covariance, a rule may use a preferred coordinate system, coding, boundary chart, or presentation convention rather than the unitary-equivalence class of T .*

Thus the order-theoretic endpoint condition is exactly form dominance; covariance is the accompanying invariance condition that prevents the selector from being a presentation artifact.

Proof. For (i), the Krein–von Neumann extension is characterized by a unitary-invariant closed-form extremal property. Therefore, if $T' = WTW^{-1}$, the Krein extension of T' is WT_KW^{-1} . The displayed rule is consequently well defined and unitarily covariant. When $T_K \neq T_F$, it is a covariant non-Friedrichs selector, so covariance alone does not determine the Friedrichs extension. The rule fails precisely the form-dominance condition.

For (ii), form dominance already puts the selected extension at the unique greatest point of the closed-form order, which is T_F . Dropping covariance does not change that pointwise conclusion, but it changes the status of the rule: a selection depending on an arbitrary chart, coding, basis, or presentation does not commute with unitary redescription and is not a canonical selector on the operator-theoretic data. This proves the stated sharpness. $\square$

Corollary 7.17 (Friedrichs–Krein endpoint distinction). *Let $T \geq 0$ be densely defined and symmetric. In the closed-form order on the nonnegative self-adjoint extensions of T , the Friedrichs extension is the greatest element and the*

Krein–von Neumann extension is the least element. Consequently, the Krein–von Neumann extension is not a counterexample to Theorem 7.15: it satisfies the dual form-minimal endpoint criterion and violates form dominance unless the two endpoints coincide.

Proof. The Friedrichs statement is the extremal characterization used in Theorem 7.15. The Krein–von Neumann statement is the dual classical extremal characterization of the minimal nonnegative self-adjoint extension. If T_K were also form-dominant, then $T_K \preceq_q T_F$ and $T_F \preceq_q T_K$ would both hold, so antisymmetry would imply $T_K = T_F$. Thus, when the endpoints are distinct, the Krein extension is selected only after the opposite endpoint criterion is added. $\square$

Corollary 7.18 (Friedrichs selection as the surviving form-dominant selector). *For a semibounded symmetric operator, the symmetric operator graph alone does not select a unique self-adjoint realization in the equal nonzero deficiency case. If the closed-form endpoint condition of form dominance is included, the unique covariant selection is the Friedrichs extension. If the dual endpoint condition of form minimality is included in the nonnegative case, the corresponding canonical selection is the Krein–von Neumann extension. Outside these explicit endpoint criteria, a particular non-unique self-adjoint extension requires boundary, phase, normalization, comparison, variational, or physical/evaluative data.*

Corollary 7.19 (Deficiency-space parameter as auxiliary operator data). *Under the hypotheses of Theorem 7.13, the von Neumann unitary $U : \mathcal{N}_+ \rightarrow \mathcal{N}_-$ has exactly one of the following fixed-base statuses:*

- (i) *no unitary intertwiner between the deficiency representations exists, so no admissibility-invariant internal self-adjoint extension selection is available;*
- (ii) *invariant unitary intertwiners exist, in which case their scalar phase orbits consist of distinct invariant extension parameters, so a unique choice still requires phase, boundary, normalization, comparison, or other selection data;*
- (iii) *U is chosen from $\mathfrak{U}(T)$ without being determined by the original base, in which case U is auxiliary boundary or deficiency-space data external to $\mathcal{B}_T$;*
- (iv) *the selected T_U is adopted as the operative observable, in which case the admissible operator domain and the operator framework have changed.*

Thus non-unique self-adjoint extension is not merely classified after the fact: the fixed-base criterion imposes a concrete intertwining condition and then a phase-orbit non-uniqueness obstruction on any claimed internal unique selection of an extension parameter.

Corollary 7.20 (Essential self-adjointness as the conservative limiting case). *If T is essentially self-adjoint, then the closure $\overline{T}$ is uniquely fixed by the original operator graph together with the Hilbert-space structure. In that*

case the self-adjoint realization introduces no choice among distinct extension parameters. Its old-domain readback is conservative. Any additional claim concerning vectors outside the original domain belongs to the closed operator framework rather than to new same-base operator content on the original partial operator base.

Corollary 7.21 (Boundary conditions as external operator data). *When a symmetric operator admits multiple self-adjoint extensions, the choice among them is not fixed by symmetry alone. Boundary conditions or equivalent extension parameters perform load-bearing work. Relative to the original symmetric operator base, such data fall under the auxiliary-data locus of the five-locus exhaustion theorem.*

Theorem 7.22 (Flagship case-study classification). *Each of the flagship families in this section, including the concrete self-adjoint extension instance, is governed by the same fixed-base dichotomy. A purported strengthening is either conservative on old-base-relevant content because the load-bearing datum is already internally fixed, or else it depends on external data, witnesses, evaluators, boundary conditions, extension parameters, or ambient extension and is not internal to the original base. In the quotient, completion, and self-adjoint extension cases, the move may fail carrier preservation outright by changing the object or operator framework under discussion.*

Proof. Combine Propositions 7.1, 7.3, 7.6, 7.7, 7.9, Theorems 7.11, 7.13, and 7.15. $\square$

8. NO EMERGENT INTERNAL POWER UNDER COMBINATION

After the family-by-family analysis, the remaining loophole is interaction. One can ask whether combinations of already classified moves—composition, iteration, diagonal packaging, or representation change—produce a genuinely new internal mechanism. They do not. Interaction can increase mathematical reach, but it does not create a third fixed-base status beyond conservative readback and external dependence.

Definition 8.1 (Interaction-generated closure). Let $\mathcal{F}_0$ be the class of purported carrier-preserving operators on a fixed base structure $\mathcal{B}$ already classified by Proposition 6.4 and Theorem 7.22. The *interaction-generated closure* $\langle \mathcal{F}_0 \rangle$ is the smallest class of old-base-relevant procedures containing $\mathcal{F}_0$ and closed under:

- (i) finite composition;
- (ii) finite iteration;
- (iii) admissibility-preserving coding, diagonal packaging, or fixed-point presentation of an already classified procedure;
- (iv) admissibility-preserving representation change, intertranslation, or temporary foundation change followed by readback to old-base-relevant content.

Lemma 8.2 (Conservative closure under composition). *If F and G each act conservatively on the old-base-relevant content of a fixed base structure $\mathcal{B}$, then $G \circ F$ acts conservatively on that same content.*

Proof. An operator that acts conservatively on old-base-relevant content adds no new values for old operations on old tuples and no new old-base consequences. Composing two such operators preserves that status. $\square$

Theorem 8.3 (Composition and iteration do not create a third case). *Let $F_1, \dots, F_n$ be procedures on a fixed base structure $\mathcal{B}$, each already classified as conservative on old-base-relevant content or external to $\mathcal{B}$. Then the composite*

$$F_n \circ \dots \circ F_1$$

has old-base-relevant effect of exactly one of the following kinds:

- (a) *it is conservative on the old base;*
- (b) *it depends essentially on an external parameter, witness, evaluator, or ambient enlargement and is external to the old base.*

Proof. Proceed by induction on n . The case $n = 1$ is the base classification. If the $(n - 1)$ -stage composite is conservative and F_n is conservative, the full composite is conservative by Lemma 8.2. The interaction closure theorem reduces any non-conservative composite to the earliest stage at which external dependence arises. If an earlier stage is externally dependent and the final old-base effect still depends on that load-bearing datum, no later stage makes the datum internally fixed; the full composite remains external by Theorem 5.3 and Corollary 5.4. The remaining possibility is that an external detour is used only as a proving environment and the final readback no longer depends on the external witness as content. Then the effect of the total composite is conservative rather than internally generative. $\square$

Proposition 8.4 (Diagonalization and fixed-point packaging). *Let D be a purported carrier-preserving operator on a fixed base structure $\mathcal{B}$ whose presentation proceeds by coding, self-application, diagonalization, or fixed-point packaging. Then exactly one of the following holds.*

- (a) *The relevant coding, substitution, and readback apparatus are already internally fixed by $\mathcal{B}$. Then adjoining D amounts only to conservative definitional expansion or admissible repackaging.*
- (b) *The construction depends essentially on an external satisfaction device, evaluation rule, witness structure, or ambient semantic framework not internally fixed by $\mathcal{B}$. Then D is external to $\mathcal{B}$.*

Proof. If the coding and substitution maps are fixed by the base, the purported diagonal operator repackages material already available in the original regime. If stronger evaluative apparatus is required, the operator depends on auxiliary data not fixed by the old base and is external by Theorem 5.3. Self-reference also does not bypass Theorem 4.6: if diagonalization makes an old partial operation newly defined on an old tuple, it changes the old graph. $\square$

Proposition 8.5 (Representation change and foundation swap). *Let T be a procedure that changes notation, coding, signature, presentation, or ambient foundational background and then reads results back to a fixed original base $\mathcal{B}$. Then exactly one of the following holds.*

- (a) *The translation preserves and reflects the old-base-relevant content of $\mathcal{B}$. Then the procedure is conservative on that content.*
- (b) *The translation changes the admissibility regime, permitted witnesses, carrier under discussion, or ambient existence assumptions. Then it is an explicit scope change and not an internal strengthening of the original base.*

Proof. If the translation preserves and reflects old-base-relevant content, it is a change of presentation. It may simplify proofs or clarify dependencies, but it does not add new old-tuple values or old-base consequences. If instead it changes admissibility, available witnesses, ambient objects, or the carrier, the procedure is no longer measured against the original regime alone. Reading a conclusion back may still yield a conservative theorem about that base, but the route is external rather than internally generative. $\square$

Theorem 8.6 (No-emergent-family theorem). *Let $\mathcal{B}$ be a fixed base structure, and let $F \in \langle \mathcal{F}_0 \rangle$ be any old-base-relevant procedure obtained from the already classified families by finite composition, iteration, diagonal packaging, or representation/foundation detour with readback. Then the effect of F on the old base is either conservative or external. Interaction among the classified families does not generate a new source of same-base operator content on $\mathcal{B}$.*

Proof. Every generator in $\mathcal{F}_0$ is already classified by Proposition 6.4 and Theorem 7.22. Theorem 8.3 handles finite composition and iteration, Proposition 8.4 handles diagonal and fixed-point packaging, and Proposition 8.5 handles coding, translation, and foundation change. $\square$

9. MAIN THEOREM AND FOUNDATIONAL CONSEQUENCE

The preceding sections provide the full chain of argument. Theorem 4.6 blocks direct enlargement of old primitive graphs on old tuples. Theorem 5.3 classifies purported operators by the status of their load-bearing auxiliary data. Theorem 6.3 and Proposition 6.4 show that the standard apparent strengthening mechanisms are exhausted and disposed of. Theorem 7.22 shows that the same dichotomy governs the flagship families. Theorem 8.6 closes the interaction loophole.

Non-definitional status of the classification. The main theorem is a direct consequence of the structural exhaustion result. The conservative/external conclusion is not stipulated. The rigidity theorem in Section 4 independently forbids alteration of old primitive graphs under a carrier-preserving extension, while the five-locus exhaustion theorem in Section 6 shows that any apparent strengthening must operate by old-graph change, carrier change, auxiliary

evaluator or witness dependence, language-only redescription, or admissibility-policy change. The auxiliary-data dichotomy then classifies these structurally constrained cases. Hence the theorem derives the classification from fixed-base constraints rather than encoding it in the definitions of “internal” or “external”.

Theorem 9.1 (Fixed-base classification theorem). *Let $\mathcal{B} = (X, \Omega, \mathcal{A})$ be a nondegenerate base structure. Let F be a purported carrier-preserving operator on $\mathcal{B}$ belonging to the interaction-generated closure of the strengthening mechanisms classified in Sections 6–8. Then exactly one of the following holds.*

- (a) *F acts conservatively on the old-base-relevant content of $\mathcal{B}$.*
- (b) *F depends essentially on external parameters, witnesses, evaluators, or ambient enlargement and is not internal to $\mathcal{B}$.*

Consequently:

- (i) *no such operator totalizes a genuinely partial old primitive on old tuples;*
- (ii) *no composition, iteration, diagonal packaging, or representation change among the classified families yields a new carrier-preserving source of admissible operator power on the original base.*

Proof. By Theorem 6.3, every apparent internal strengthening of the fixed base falls under one of the five loci in Definition 6.2. Proposition 6.4 shows that each locus is blocked, conservative, external, or a scope change. Theorem 7.22 applies that classification to the flagship families. Theorem 8.6 shows that the same classification is stable under the natural interaction operations by which those families might be recombined. The two alternatives collect the conservative cases and the externally dependent cases. Consequence (i) is Theorem 4.6; consequence (ii) is Theorem 8.6. $\square$

Structural rigidity. The classification result also gives a rigidity property. Operator-level transformations decompose, relative to the fixed base, into a structure-reducing or scope-changing component and a structure-preserving component. A transformation that preserves all structural invariants of the fixed base is therefore equivalent to the identity on old-base-relevant content, up to representation. Nontrivial strength enters only by changing carrier, changing admissibility policy, or introducing comparison data not fixed by the base.

Operator-family consequences. The structural classification yields a disciplined comparison language for standard constructions. Procedures based on completion, selection, canonical representation, quotienting, universal characterization, compactness, saturation, or forcing-style enrichment either introduce additional structure, such as enlarged domains or comparison frameworks, or reduce to conservative reformulations of content already fixed by the base.

Corollary 9.2 (No internal re-entry principle). *External methods may establish theorems, transfer principles, canonical readbacks, and invariance claims about a fixed base. What they do not do is return as new same-base operator content on that base unless the return map is conservative on old-base-relevant content.*

Proof. Immediate from Theorem 9.1 and Theorem 8.6. $\square$

The intended consequence is exact rather than rhetorical. The paper does not say that external methods are dispensable, that their outputs are merely verbal, or that all mathematically important distinctions reduce to definitional equivalence. It says something narrower. Once a base structure and its admissibility regime are held fixed, the fixed-base content returned by those methods must be conservative unless the additional structure used by the method is explicitly included in the comparison.

10. OPERATOR-THEORETIC CONSEQUENCES FOR QUANTUM FRAMEWORKS

The preceding results are mathematical fixed-base theorems. This section records their consequences for operator-theoretic formulations of quantum mechanics. No modification of quantum mechanics, no new empirical prediction, and no alternative physical dynamics are asserted. The claim is structural: once a state carrier, operator family, admissibility regime, and readback language are fixed, operator-level extension and readback cannot be treated as new same-base operator content unless its old-base readback is conservative.

Definition 10.1 (Quantum operator base). A quantum operator base is a structure

$$\mathcal{B}_{\text{qm}} = (S, \Omega_{\text{qm}}, \mathcal{A}_{\text{qm}}),$$

where S is a fixed state carrier or representation space, Ω_{qm} is a family of partial or total operators representing observables, evolutions, constraints, measurements, or comparison maps, and $\mathcal{A}_{\text{qm}}$ specifies the admissible domains, compositions, and readback rules licensed in the framework.

The carrier S may be a Hilbert space, a dense invariant domain, a space of density operators, an algebraic state space, or another representation space. The operator family may include unbounded observables, self-adjoint operators, spectral projections, POVMs, dynamics, constraint operators, or expectation maps, as in standard operator-algebraic and functional-analytic treatments of quantum theory [5, 13, 15]. The theorem applies only after these objects and their readback language have been fixed.

Theorem 10.2 (Fixed-base constraint for quantum operator frameworks). *Let $\mathcal{B}_{\text{qm}} = (S, \Omega_{\text{qm}}, \mathcal{A}_{\text{qm}})$ be a fixed quantum operator base. Any operator-level extension and readback of $\mathcal{B}_{\text{qm}}$ has old-base-relevant effect of exactly one of the following kinds:*

- (i) conservative reformulation of operator content already fixed on the original base;
- (ii) alteration of the old operator graph or old admissible domain;
- (iii) change of state carrier, representation space, quotient, completion, or ambient operator framework;
- (iv) dependence on added evaluative, boundary, comparison, measurement, or readback data;
- (v) language-only expansion;
- (vi) change of admissibility policy or operator family.

In particular, no strengthening yields new operator power on the fixed quantum operator base unless its returned content is conservative or its dependence on additional structure is made explicit.

Proof. This is Theorem 9.1 applied to $\mathcal{B}_{\text{qm}}$. The state carrier, operator family, admissibility regime, and readback language play the roles of the old carrier, old partial operations, admissibility profile, and old-base-relevant comparison language. $\square$

10.1. Domain and self-adjoint extension constraint. For unbounded quantum observables, domain data are not cosmetic. A symmetric operator, its closure, and its self-adjoint extensions may agree on an initial domain while differing on admissible vectors, boundary conditions, spectral content, and physical interpretation. The fixed-base theorem imposes the following constraint.

Corollary 10.3 (Domain-extension constraint). *Let T be a densely defined symmetric operator in a fixed quantum operator base. Passing to a self-adjoint extension $\tilde{T}$ is one of the following:*

- (i) conservative readback to the original domain;
- (ii) a change of admissible operator domain;
- (iii) dependence on boundary or deficiency-space data not fixed by the original base;
- (iv) replacement of the original operator framework by a new self-adjoint framework.

It is not a same-base internal totalization of the original partial operator.

This is the concrete quantum-mechanical instance of Theorem 4.6 developed in Theorem 7.11. If an operator application was undefined on the original domain, later defining it requires either changing the operator graph, changing the admissibility policy, or moving to a new operator framework.

Corollary 10.4 (Deficiency-index invariant-selection constraint). *Suppose a closed symmetric operator T has equal nonzero deficiency indices. Let G_T be the unitary redescription group of the fixed symmetric-operator base, and let $\rho_{\pm}$ be its induced representations on $\mathcal{N}_{\pm}$. Then fixed-base internal extension candidates are exactly the unitary intertwiners*

$$\text{UHom}_{G_T}(\mathcal{N}_+, \mathcal{N}_-).$$

If this set is empty, the original quantum operator base contains no admissibility-invariant internal selector of a self-adjoint realization. If it is nonempty, each candidate generates a scalar phase family of pairwise distinct self-adjoint realizations. Therefore the original base still does not determine a unique realization unless additional boundary, phase, normalization, comparison, variational, or physical/evaluative data are included in the base. Consequently, any claim depending on a particular self-adjoint realization must identify the operator-framework data that select that realization.

Corollary 10.5 (Friedrichs endpoint characterization constraint). *Let T be a lower semibounded densely defined closed symmetric operator in a fixed quantum operator base. If the base includes the closed quadratic-form order on $\mathcal{E}_m(T)$ and imposes form dominance as the endpoint criterion, then the unique selected self-adjoint realization is the Friedrichs extension T_F . If the opposite endpoint criterion is imposed in the nonnegative case, the selected endpoint is the Krein–von Neumann extension when it is defined. If neither endpoint criterion is included, invoking a particular extension adds variational, boundary, form-order, or physical selection data to the operator framework. Thus Friedrichs selection is the surviving canonical case precisely because the form-dominance criterion is explicitly part of the base.*

10.2. Spectral and representation constraints. Spectral constructions often pass from an operator to a spectral measure, functional calculus, direct-integral representation, rigged Hilbert-space setting, or equivalent representation. These constructions are mathematically legitimate. Their fixed-base status is determined by what performs the load-bearing work.

Corollary 10.6 (Spectral representation constraint). *Let a spectral or representation-theoretic construction be used to strengthen a fixed quantum operator base. If the construction changes the representation space, completion, spectral environment, or functional-calculus setting, then it is a scope change. If its readback to the original operator base is already fixed by the original data, the readback is conservative. If the readback depends on an added comparison map, boundary convention, representation choice, or evaluator, the construction depends on external auxiliary data.*

Thus spectral representation does not fail; it is classified. The theorem separates valid use of spectral machinery from the stronger claim that such machinery creates new same-base operator content on an unchanged base.

10.3. Measurement and evaluative-structure constraint. Measurement formalisms require evaluative structure: probability rules, expectation maps, spectral measures, POVMs, pointer readbacks, coarse-grainings, or outcome-comparison criteria. In the fixed-base analysis, these are not invisible.

Corollary 10.7 (Measurement-readback constraint). *Let a measurement or observable scheme assign outcomes, probabilities, or expectation values relative to a fixed quantum operator base. If the evaluative structure is part of*

the original base, the readback is internal to that framework. If it is introduced to resolve an otherwise underdetermined operator-level question, it is auxiliary data. The resulting construction is therefore external to the original base unless its old-base effect is conservative.

This does not challenge the role of measurement rules in quantum mechanics. It states the fixed-base bookkeeping condition: measurement data must either be included in the operator base or counted as added evaluative structure.

10.4. Quantum–classical interface constraint. At the quantum–classical interface, one compares frameworks with different state carriers, observable algebras, admissibility regimes, or readback maps. Decoherence, coarse-graining, semiclassical limits, classical observables, and representation changes may all be mathematically and physically indispensable [16]. The fixed-base theorem constrains how such transitions are classified.

Corollary 10.8 (Quantum–classical interface constraint). *A passage between quantum and classical operator frameworks is, relative to a fixed starting base, one of the following:*

- (i) *conservative readback inside the starting framework;*
- (ii) *change of carrier, algebra, state space, or representation;*
- (iii) *introduction of coarse-graining, pointer, probability, or comparison data;*
- (iv) *language-only redescription;*
- (v) *change of admissibility policy.*

It cannot be treated as new same-base operator content on a single unchanged base unless the carrier, operator graph, admissibility regime, and readback language remain fixed and the returned content is conservative.

The result is therefore not a reduction of quantum theory to classical theory, or classical theory to quantum theory. It is a structural constraint on operator-framework comparison: interface constructions must identify whether their strength comes from conservative readback, carrier change, auxiliary evaluation, or altered admissibility.

10.5. Summary of the physics-facing consequence. For quantum-mechanical operator frameworks, the theorem yields a domain and readback discipline:

operator extension and readback

$\Rightarrow$

conservative readback, external data, or changed framework.

This is the relevant mathematical-physics consequence. Operator-theoretic methods such as self-adjoint extension, Friedrichs endpoint characterization, completion, spectral representation, measurement readback, and quantum–classical comparison remain valid. The theorem specifies when they remain

internal to a fixed base and when they derive their strength from additional operator, boundary, variational/form, representational, or evaluative structure.

11. CONCLUSION

The paper proves a fixed-base classification theorem for operator-level extension and readback mechanisms, with self-adjoint extension selection as the main operator-theoretic instance. Relative to a fixed base structure equipped with an admissibility regime, purported carrier-preserving operator power on old-base-relevant content is either conservative or else depends on external witnesses, evaluators, comparison data, boundary conditions, deficiency-space parameters, or ambient enlargement. In particular, no carrier-preserving extension totalizes a genuinely partial old primitive on old tuples, and no recombination of the standard detours considered here yields a third internal case.

The concrete operator-theoretic instance is self-adjoint extension of symmetric operators. When the deficiency indices are equal and nonzero, the von Neumann unitary between deficiency subspaces is the load-bearing extension datum. The invariant deficiency-parameter and invariant-selection theorems sharpen this point: redescription-invariant candidates are exactly unitary intertwiners between the deficiency representations, and any such candidate generates a scalar phase family of pairwise distinct self-adjoint extensions. The Friedrichs endpoint characterization theorem supplies the positive canonical endpoint: for a lower semibounded closed symmetric operator, the upper endpoint condition in the closed quadratic-form order is necessary and sufficient for selecting the Friedrichs extension, while the Krein–von Neumann extension is obtained only by the opposite lower-endpoint criterion. Essential self-adjointness is therefore the conservative limiting case; Friedrichs or Krein selection is the explicit form-order endpoint case; all remaining non-unique self-adjoint extension requires boundary, phase, normalization, comparison, variational, or physical/evaluative data unless such data are already included in the base.

That conclusion leaves full room for the mathematical and physical strength of completion, forcing, saturation, monster-model arguments, spectral constructions, representation changes, and universal constructions. Their role is not denied but classified: they are external or scope-extending methods unless their results are brought back to the original base by a conservative readback. The significance of the result lies in separating conservative readback, auxiliary extension data, and changed operator framework, with the invariant-selection theorem and Friedrichs endpoint characterization giving the standard quantum-mechanical model cases. For mathematical physics, this supplies a precise constraint on when operator-theoretic extensions of a fixed state or observable framework remain internal and when they require

additional representational, boundary, deficiency-space, variational/form, or evaluative support.

APPENDIX A. EXTENDED FAMILY PROOFS

The main text isolates, for each flagship family, the load-bearing datum or ambient move and then applies the formal spine developed in Sections 4–6. This appendix records longer reductions and mixed-family examples. It introduces no new theorem family; it makes explicit how the main classification works inside recurring subcases.

A.1. Selection rules and quotients.

Proposition A.1 (Representative rules factor through auxiliary data). *Let $D \subseteq X$ be admissible in $\mathcal{B} = (X, \Omega, \mathcal{A})$, and let $\sim$ be an admissibility-invariant equivalence relation on D . Suppose a purported carrier-preserving normalization $N : D \rightarrow D$ is constant on $\sim$ -classes and satisfies $N(x) \sim x$ for all $x \in D$. Then the fixed-base status of N is determined by the status of the representative rule*

$$[x]_{\sim} \mapsto N(x).$$

If that rule is internally fixed, N is conservative on old-base-relevant content; if not, N is external. If one passes instead to the quotient $D/\sim$, the move is not carrier-preserving.

Proof. Because N is constant on classes, its effect factors through the quotient map together with a choice of section on the relevant quotient classes. The section is the load-bearing datum. Theorem 5.3 classifies the section as internally fixed or external. Replacement of each element by its equivalence class is a carrier change and is covered by Proposition 6.4. $\square$

A.2. Completion arguments as enlargement plus readback.

Proposition A.2 (Completion decomposes into enlargement and readback). *Let A be an admissible object in a fixed base structure, let $i : A \hookrightarrow \hat{A}$ be a completion-style embedding into a richer object, and let R be a procedure that reads old-base-relevant consequences about A from work carried out in $\hat{A}$. Then exactly one of the following occurs.*

- (a) *The substantive output concerns new points, ideal elements, or totalized operations in $\hat{A}$. Then the move is external by carrier change.*
- (b) *The output concerns only the old base A , and the readback map R is already internally fixed. Then the move is conservative on old-base-relevant content.*
- (c) *The output concerns only the old base A , but the readback depends on an evaluator, boundary convention, embedding policy, or comparison rule not internally fixed by A . Then the move is external.*

Proof. The completion step is passage to a richer object. If the final content remains about that richer object, it is external to the original base. If the final content is read back to the old base, the classification turns on whether the embedding and readback rule are internally fixed. This is Theorem 5.3 applied to the readback datum. $\square$

A.3. Compactness and saturation as witness import.

Proposition A.3 (Witness-import schema for model-theoretic arguments). *Let M be a structure, let $a \in M$ be old parameters, and suppose a purported operator on M is justified by passing to a larger structure $N \succeq M$, by realizing a type over M , or by invoking a compactness-based existence theorem. Then the fixed-base effect is classified as follows.*

- (a) *If the relevant witness or realization is already definably fixed in M , the move is conservative.*
- (b) *If the argument requires a new element of N , a witness model, or an existence theorem whose witness is not part of the original base, the move is external.*
- (c) *If the larger structure is used only as a detour to prove an invariant old-language statement about M , the readback is conservative even though the proving route is external.*

Proof. A saturation or compactness argument has an ambient witness layer and a final statement about original parameters. If the witness is already present and fixed in M , the ambient move adds no old-base content. If the witness lies outside M , the operator is not internal to M . If the ambient structure is used only as a proving environment for an old-language consequence, the old-base effect is conservative. $\square$

A.4. Forcing as extension plus absoluteness readback.

Proposition A.4 (Forcing decomposes into witness import and readback). *Let M be a fixed set-theoretic base, and suppose a purported operator on M is presented via forcing. Then the analysis separates into two stages.*

- (a) *The passage from M to a forcing extension $M[G]$ is external, since the generic object G is not internally fixed by M .*
- (b) *Any later conclusion about old parameters from M is either a conservative readback, when obtained by absoluteness or invariance, or else still depends on the external extension and remains external.*

Consequently forcing never supplies a third fixed-base status beyond external extension and conservative readback.

Proof. The generic filter or generic real is the load-bearing witness of the forcing argument. Because it is not part of the original base, the move to $M[G]$ is external. A transported statement about old parameters is conservative exactly when its content no longer depends on G . $\square$

A.5. Universal properties and ambient comparison data.

Proposition A.5 (Universal witnesses depend on ambient comparison data). *Let A be an object presented by a fixed base structure, and suppose an associated construction is specified by a universal property in an ambient category $\mathcal{C}$. Then the fixed-base status of the construction is carried by the ambient comparison data: the class of comparison maps, the universal condition, and the witness object satisfying it. If those data are used only to justify a conservative reformulation of old-base content, the effect is conservative; otherwise the move is external.*

Proof. A universal property quantifies over a wider ambient setting and singles out a witness object by its behavior relative to comparison maps. If the only old-base consequence is an invariant reformulation already determined there, the effect is conservative. In the substantive case, the universal witness or comparison condition does real mathematical work and is external ambient data. $\square$

A.6. Mixed-family worked examples.

Example A.6 (Completion followed by canonicalization). One first passes from an old object to a completion, then chooses a canonical representative there, and finally reads a statement back to the old base. The completion stage is external. If the representative rule used in the completed object is not internally fixed even relative to that ambient setting, a second external dependence has been added. If the final readback states only an invariant fact about the original object, the net old-base effect is conservative; otherwise the procedure remains external.

Example A.7 (Forcing followed by absoluteness). One passes from M to $M[G]$, proves a statement about old parameters, and then invokes absoluteness to transfer the statement back to M . The first stage imports the generic witness and is external. The final transfer is conservative on old-base-relevant content precisely when the transported statement no longer depends on G as part of its content.

Example A.8 (Monster-model detour followed by quotient or canonical parameter). One works in a monster model to compare types or define a quotient-like invariant and then states a theorem about the original small model. The monster model is an ambient enlargement, and any quotient passage or representative choice must still be classified by the corresponding family analysis. The old-base effect is either a conservative theorem about the small model or an external dependence on the larger ambient setting.

APPENDIX B. REFEREE-FACING DEPENDENCY MAP

This appendix isolates the proof spine. Appendix A supplies expanded applications but introduces no additional premises. Table 2 gives the main dependency chain.

TABLE 2. Dependency spine of the proof

Step	Role
D1	Section 3 fixes the base structure, old-base-relevant content, fixed comparison class, and internal/external terminology.
D2	Section 4 proves old-graph rigidity and no internal totalization.
D3	Section 5 proves the auxiliary-data dichotomy by redescription invariance.
D4	Section 6 proves the five-locus exhaustion and disposes of each locus.
D5	Section 7 applies the spine to the flagship families.
D6	Section 8 proves closure under recombination.
D7	Section 9 combines D2–D6 into the final fixed-base theorem.

Proof-obligation checklist. Table 3 summarizes the main proof obligations and where they are discharged.

TABLE 3. Proof obligations and discharge locations

Obligation	Discharge	Location
Show the theorem is not definitional	Old-graph rigidity and five-locus exhaustion are proved before the final internal/external classification is applied.	Sections 4, 6
Specify the comparison class	Carrier, old graphs, admissibility profile, and old comparison language are fixed explicitly.	Section 3
Cover apparent strengthening routes	Every old-base-relevant change must enter one of five structural loci.	Theorem 6.3
Handle hidden witnesses and semantic relabeling	Redescription invariance distinguishes internally fixed data from external data.	Section 5
Avoid overclaiming about forcing/model theory	The paper distinguishes external method from conservative old-base readback.	Propositions 7.6, 7.7
Close recombination loop-holes	Composition, iteration, diagonal packaging, and representation detours preserve the conservative/external dichotomy.	Section 8

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